Descriptive Statistics

Chapter 3

Biostatistics and Data Analysis

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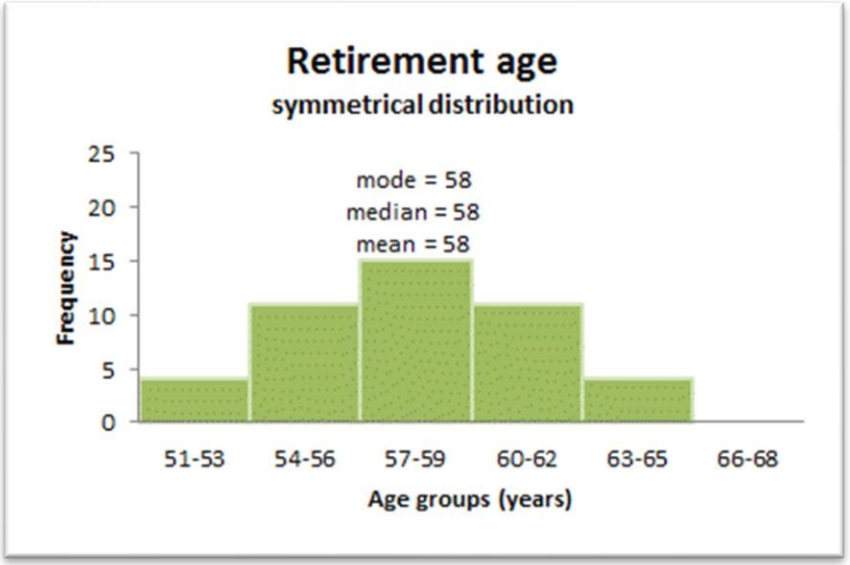
Descriptive statistics

- Descriptive statistics results from gathering data from a body, group, or population and reaching conclusions only about that particular group or population.
- **For example**
  - ✓ If an instructor produces statistics to summarize a class's examination effort and uses those statistics to reach conclusions about that class only, the statistics are descriptive.

Descriptive Statistics:

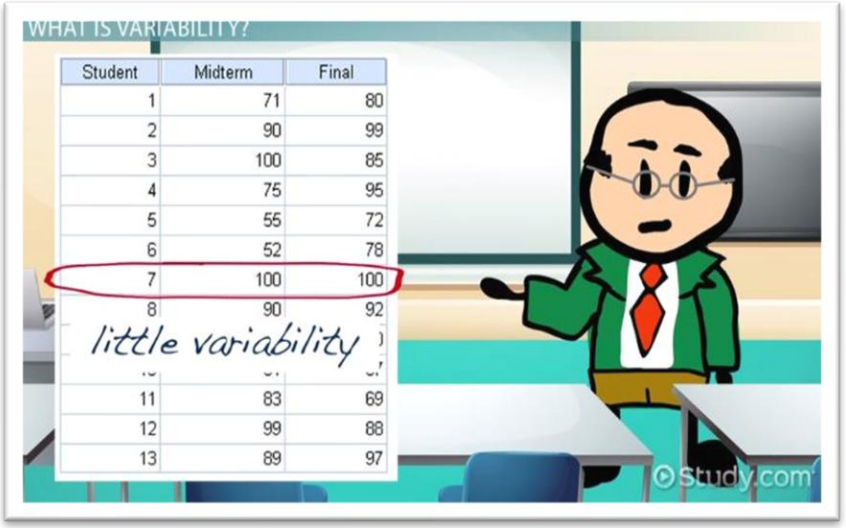
Measures of central tendency

- In statistics, the central tendency reflects the centre of the data distribution.
- There are three common measures of central tendency:
  - Mode (most frequently occurring number)
  - Median (middle, midpoint)
  - Mean (average)



Measures of Variability

- Variability measure how much your scores differ from each other.
  - Variance (how far the numbers are spread out)
  - Standard deviation (how much variation from the mean/average)
  - Range (the difference between the largest and smallest variable)



Measures of Central Tendency – *Mean*

- Arithmetic mean is a mathematical average and it is the most popular measures of central tendency.

- It is frequently referred to as 'mean' it is obtained by dividing sum of the values of all observations in a series (ΣX) by the number of items (n) constituting the series.

- Thus, mean of a set of numbers X1, X2, X3,……..Xn denoted by $\bar{x}$ and is defined as

$$\bar{x} = \frac{\sum x}{n} = \frac{the\ \ sum\ \ of\ \ the\ \ data\ \ pts}{the\ \ number\ \ data\ \ pts}$$

Measures of Central Tendency – ***Mean***

$$\bar{x} = \frac{\sum x}{n} = \frac{the \quad sum \quad of \quad the \quad data \quad pts}{the \quad number \quad data \quad pts}$$

$\bar{x}$ …is the sample mean (read "x bar")

n … is the number of sample observations

x … is a particular value

Σ … indicates the operation of adding (sigma)

Measures of Central Tendency – *Mean*

Example I

Additional Example 1A: Finding the Mean of a Data Set



Find the mean of each data set.

Depths of Puddles (in.)						
5	8	3	5	4	2	1

Mean

5 + 8 + 3 + 5 + 4 + 2 + 1 = 28 *(Add all values)*

28 ÷ 7 = 4 *(Divide the sum by the number of items)*

The mean is 4 inches.

Measures of Central Tendency – *Mean*
Example 2

Additional Example 1B: Finding the Mean of a Data Set

Find the mean of each data set.

Number of Points Scored			
96	75	84	7

Mean:

96 + 75 + 84 + 7 = 262 *Add all values.*

262 ÷ 4 = 65.5 *Divide the sum by the number of items.*

The mean is 65.5 points. The average number of points scored is 65.5.

Measures of Central Tendency – *Mean*

Example 3

Check It Out: Example 1A

Find the mean of each data set.

Rainfall per month (in.)						
1	2	10	2	5	6	9

$$\frac{1 + 2 + 10 + 2 + 5 + 6 + 9}{7}$$

$$= \frac{35}{7}$$

$$= 5$$

The mean is 5 in.

Measures of Central Tendency – *Mean*
Example 4

Check It Out: Example 1B

Find the mean of each data set.

82	59	43	75	57	59

$$\frac{82 + 59 + 43 + 75 + 57 + 59}{6}$$

$$= \frac{375}{6}$$

$$= 62.5$$

Measures of Central Tendency – *Mean*

Example 5

A sample of five executives received the following bonuses last year :

14.0 15.0 17.0 16.0 15.0

Determine the average bonus given last year:

Formula $\bar{x} = \frac{\sum x}{n}$

$= \frac{14 + 15 + 17 + 16 + 15}{5}$

$= 77 / 5 \qquad = 15.4$

Measures of Central Tendency – *Mean*

- **Example 6**
  - Data: 8, 4, 2, 6, 10

$$\bar{x} = \frac{(8+4+2+6+10)}{5} = 6$$

- **Example 7**
  - Sample: 10 trees randomly selected from Battle Park
  - Diameter (inches):

    9.8, 10.2, 10.1, 14.5, 17.5, 13.9, 20.0, 15.5, 7.8, 24.5

$$\bar{x} = \frac{(9.8+10.2+\cdots+24.5)}{10} = 14.38$$

Example 8: Calculated the Arithmetic ***Mean*** Monthly Users Statistics in the University Library

Month	No. of Working Days	Total Users	Average Users per month
Sep-2011	24	11618	
Oct-2011	21	8857	
Nov-2011	23	11459	
Dec-2011	25	8841	
Jan-2012	24	5478	
Feb-2012	23	10811	
Total			

$$\text{Mean} = \frac{\text{Total number of users}}{\text{Total number of working days}}$$

$$= \frac{57064}{140}$$

$$= \mathbf{407.6}$$

Example 8: Calculated the Arithmetic Mean Monthly Users Statistics in the University Library

Month	No. of Working Days	Total Users	Average Users per month
Sep-2011	24	11618	484.08
Oct-2011	21	8857	421.76
Nov-2011	23	11459	498.22
Dec-2011	25	8841	353.64
Jan-2012	24	5478	228.25
Feb-2012	23	10811	470.04
Total	**140**	**57064**	

$$\text{Mean} = \frac{\text{Total number of users}}{\text{Total number of working days}}$$

$$= \frac{57064}{140}$$

$$= 407.6$$

Measures of Central Tendency

Weighted Mean

The **weighted mean** of n numbers $x_1, x_2, \ldots, x_n$, that are weighted by the respective factors $f_1, f_2, \ldots, f_n$ is given by the formula:

$$\overline{w} = \frac{\sum (x \cdot f)}{\sum f}.$$

Measures of Central Tendency

Weighted Mean

Listed below are the grades of a students semester courses. Calculate the Grade Point Average (GPA).

Course	Grade	Points (x)	Credits (f)	x * f
Calculus	A	4	5	20
Chemistry	B	3	3	9
Physics	A	4	2	8
Biology	C	2	2	4

$$\overline{w} = \frac{\Sigma(x*f)}{\Sigma f} \qquad \overline{w} = \frac{(4*5)+(3*3)+(4*2)+(2*2)}{5+3+2+2} \qquad \overline{w} = \frac{20+9+8+4}{12}$$

$$\overline{w} = \frac{41}{12} \qquad \overline{w} = 3.42 \qquad \text{GPA} = 3.42$$

Measures of Central Tendency

Advantages of Mean:

- It is easy to understand & simple to calculate.
- It is based on all the values.
- Sensitive to changes in data
- Provides a central tendency measure
- It is easy to understand the arithmetic average even if some of the details of the data are lacking.

Disadvantages of Mean:

- It is affected by extreme values.
- It cannot be calculated for open end classes.
- It cannot be located graphically

Measures of Central Tendency

Median

- Another measure of central tendency, is the **median**.
- This measure divides a group of numbers into two parts, with half the numbers below the median and half above it.
- The median is not as sensitive to extreme values as the mean.
- To find the **median** of a group of items:

1. Rank the items.

2. If the number of items is **odd**, the median is the middle item in the list.

3. If the number of items is **even**, the median is the mean of the two middle numbers.

Measures of Central Tendency

Median

Example:

Ten students in a math class were polled as to the number of siblings in their individual families and the results were:

3, 2, 2, 1, 1, 6, 3, 3, 4, 2.

Find the median number of siblings for the ten students.

Position of the median: $10/2 = 5$

Between the 5th and 6th values

Data in order: 1, 1, 2, 2, 2, 3, 3, 3, 4, 6

Median $= (2+3)/2 = 2.5$ siblings

Measures of Central Tendency

Median

Example:

Nine students in a math class were polled as to the number of siblings in their individual families and the results were:

3, 2, 2, 1, 6, 3, 3, 4, 2.

Find the median number of siblings for the ten students.

Position of the median: 9/2 = 4.5

The 5th value

In order: 1, 2, 2, 2, 3, 3, 3, 4, 6

Median = 3 siblings

Measures of Central Tendency

Median

Median in a Frequency Distribution

Example:

Find the median for the distribution.

Value (x)	1	2	3	4	5
Frequency (f)	4	3	2	6	8

Position of the median is the sum of the frequencies divided by 2.

$$\text{Position of the median} = \frac{\Sigma (f)}{2} = \frac{23}{2} = 11.5 = 12^{th} \text{ term}$$

Add the frequencies from either side until the sum is 12.

The 12th term is the median and its value is 4.

Measures of Central Tendency

Advantages of Median:

- The median is not affected by very large or very small values.
- Median can be understood even by common people.
- Median can be determined even with the extreme values.

Disadvantages of Median:

- If there is an even number of numbers, the median is found by averaging the two middle numbers. This means the median value may not actually be a number in the original data set.

Measures of Central Tendency

Mode

- The **mode** of a data set is the value that occurs the most often.
- If a distribution has two modes, then it is called **bimodal**.
- In a large distribution, this term is commonly applied even when the two modes do not have exactly the same frequency

Example:

Ten students in a math class were polled as to the number of siblings in their individual families and the results were: 3, 2, 2, 1, 3, 6, 3, 3, 4, 2. Find the mode for the number of siblings.

3, 2, 2, 1, 3, 6, 3, 3, 4, 2

The mode for the number of siblings is 3.

Measures of Central Tendency

Mode

Mode in a Frequency Distribution

Example:

Find the mode for the distribution.

Value (x)	1	2	3	4	5
Frequency (f)	4	3	2	6	8

The mode in a frequency distribution is the value that has the largest frequency.

The mode for this frequency distribution is 5 as it occurs eight times.

Measures of Central Tendency

Mode

Mode in a Frequency Distribution

Example:

Find the mode for the distribution.

Value (x)	1	2	3	4	5
Frequency (f)	1	1	1	1	1

- Each number appears only once.
- ***Therefore, there is: No mode***
- Since all numbers have the same frequency (1), the dataset has no central tendency or most frequent value.

Measures of Central Tendency

Mode

Mode in a Frequency Distribution

Example:

Find the mode for the distribution.

Value (x)	1	2	3	4	5
Frequency (f)	1	6	6	2	3

- Two modes: 2 and 3 (both have frequency 6)
- Mode: 2 and 3
- This distribution is bimodal because there are two common values.

Measures of Central Tendency

Advantages of Mode :

- Mode is readily comprehensible and easily calculated
- It is the best representative of data
- It is not at all affected by extreme value.
- The only averages that can be used if the data set is not in numbers (Useful for categorical data).

Disadvantages of Mode :

- There can be more than one mode.
- There can also be no mode.

Measures of Central Tendency

Conclusion

- A measure of central tendency is a measure that tells us where the middle of a data lies.
- Mean is the most common measure of central tendency. It is simply the sum of the numbers divided by the number of numbers in a set of data. This is also known as average.
- Median is the number present in the middle when the numbers in a set of data are arranged in ascending or descending order. If the numbers in a data set is even, then the median is the mean of the two middle numbers.
- Mode is the value that occurs most frequently in a set of data.

Measures of Central Tendency

Additional Example : Finding the Mean, Median, Mode, and Range of a Data Set

Find the mean, median, mode, and range of the data set.

Car Wash Totals			
6th Grade	12	7th Grade	11
8th Grade	14	9th Grade	15

Mean: $\frac{12 + 11 + 14 + 15}{4} = 13$ *Add all values. Divide the sum by the number of items.*

Write the data in numerical order. 11, 12, 14, 15

median: 11, (12, 14), 15 *There are an even number of items, so find the mean of the two middle values.*

$$\frac{12 + 14}{2} = 13$$

mode: none *No value occurs most often.*

range: 15 – 11 = 4 *Subtract least value from greatest value.*

The mean is 13, the median is 13, there is no mode, and the range is 4.

Measures of Central Tendency

Example

Find the mean, median, mode, and range of each data set.

Number of Vacation Days			
53	26	47	12

mean: $\frac{53 + 26 + 47 + 12}{4} = \frac{138}{4} = 34.5$ days

median: $\frac{26 + 47}{2} = \frac{73}{2} = 36.5$ days

mode: none

range: 53 - 12 = 41 days

Measures of Central Tendency

Example

Find the mean, median, mode, and range of each data set.

Height of Tomato Plants (in.)				
6	8	11	7	8

mean: $\frac{6 + 8 + 11 + 7 + 8}{5} = \frac{40}{5}$ = 8 in.

median: 8 in.

mode: 8 in.

range: 11 - 6 = 5 in.

Measures of Central Tendency

Quiz

1. Identify the mean of the following data set.

20, 25, 60, 42, 30, 20, 20

A. 31

B. 34

C. 38

D. 42

Add numbers:20 + 25 + 60 + 42 + 30 + 20 + 20 = 217
Count numbers:7
Mean = 217 ÷ 7= 31

Measures of Central Tendency

Quiz

2. Identify the median of the following data set.

20, 22, 58, 48, 32, 20, 23

A. 48

B. 32

C. 23

D. 20

1. *Arrange numbers in order:20, 20, 22, 23, 32, 48, 581.*
2. *Count numbers:7 (odd number)*
3. *Find middle value: Since there are 7 numbers (odd), the middle value is the 4th number.*
4. *Median = 4th number= 23*

Measures of Central Tendency

Quiz

3. Identify the mode of the following data set.

20, 28, 55, 48, 30, 20, 25

A. 20

B. 23

C. 25

D. 32

1. *Count frequency of each number.*
2. *Identify most frequent value(s)*
3. *20 (appears twice)*
4. *The mode of the data set is:20*

Measures of Central Tendency

4. Rebecca spent $7, $12, $8, and $13 over the past 4 days buying vegetables. What was the mean amount spent on vegetables?

A. $10

B. $9

C. $8

D. $7

1. *Add up the daily expenses:$7 + $12 + $8 + $13 = $40.*
2. *Count the number of days: 4*
3. *Calculate the mean:*
   Mean = Total / Count= $40 / 4= $10

Measures of Position

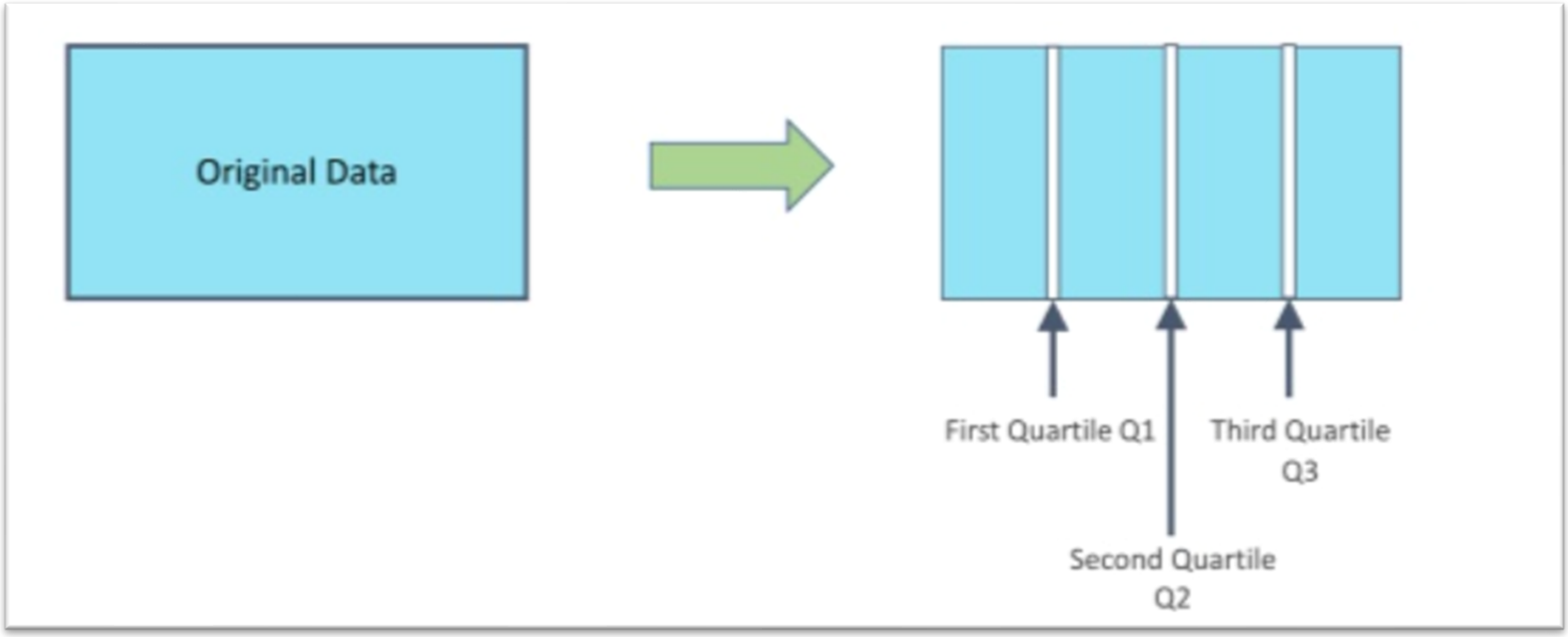
- A measure of position determines the position of a single value in relation to other values in a sample or a population data set.

- There are many measures of position, only discuss

1. Quartiles
2. Interquartile Range

Quartile

- Quartile divides a list of numbers into four quarters.
- The **three quartiles** (Q1, Q2, Q3) divide the data values at the three points that splits the data set into four equal groups.



Quartile

Types of Quartiles

1. First Quartile (Q1 or 25th Percentile)

- Lowest 25% of data
- Also known as the lower quartile
- Represents the value below which 25% of the data falls

2. Second Quartile (Q2 or 50th Percentile)

- Middle 50% of data
- Also known as the median
- Represents the middle value of the dataset

3. Third Quartile (Q3 or 75th Percentile)

- Highest 25% of data
- Also known as the upper quartile
- Represents the value above which 25% of the data falls

Quartile

How to Calculate Quartiles

1. Arrange data in ascending order.
2. Determine the number of values (n).
3. Calculate quartile positions:

- Q1: (n+1) x 25%
- Q2: (n+1) x 50%
- Q3: (n+1) x 75%

4. Find the corresponding values at these positions.

Quartile

Example: Find Quartiles Q_1, Q_2, and Q_3. 3,4,8,5,10,9,1,4,6,12,2

- Arrange the data in order: Q_2

1, 2, 3, 4, 4, **5**, 6, 8, 9, 10, 12

n = 11

Q_2 = ?

=(n + 1) *50%

=(11 +1) *0.5

=6

The number which is at 6th position is Q_2. which is 5 is median.

Quartile

- Arrange the data in order: Q_1

  1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 12

n = 11

Q_1 = ?

=(n + 1) *25%

=(11 +1) *0.25

= 3

The number which is at 3rd position is Q_1. which is 3.

- Q_3

==(n + 1) *75%

=(11 +1) *0.75

= 9

The number which is at 9th position is Q_3. which is 9.

Quartile

Example: Find Quartiles Q_1, Q_2, and Q_3. 3,4,8,5,10,9,1,4,6,12,2,14

- Arrange the data in order: Q_2

  1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 12, 14

n = 12

Q_2 = ?

=(n + 1) *50%

=(12 +1) *0.5

=6.5

The number which is at 6th & 7th position is Q_2. which is 5.5 is median.

Quartile

- Arrange the data in order: Q_1

  1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 12,14

n = 12

Q_1= ?

=(n + 1) *25%

=(12 +1) *0.25

=3.25

= The number which is at 3rd & 4th position is Q_1. $\frac{3+4}{2} = 3.5$

- Q_3

=(n + 1) *75%

=(12 +1) *0.75

= The number which is at 9th & 10th position is Q_3 $\frac{9+10}{2} = 9.5$

The interquartile range (IQR)

- The interquartile range (IQR) is the difference between the third and first quartiles: that is $Q_3 - Q_1$
- It is a measure of dispersion.
- A large IQR indicates a large amount of variability among the middle 50 percent of the relevant observations,
- A small IQR indicates a small amount of variability among the relevant observations.

$$IQR = Q3 - Q1$$

1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 12

$$IQR = 9 - 3$$

$$= 6$$

Measures of Dispersion

- A measure of dispersion conveys information regarding the amount of variability present in a set of data.

Note:

- If all the values are the same → There is no dispersion.
- If all the values are different → There is a dispersion.
- If the values close to each other →The amount of Dispersion small.
- If the values are widely scattered→ The Dispersion is greater.

Measures of Dispersion

MEASURES OF VARIABILITY

- ✓ Range
- ✓ Variance
  - Population variance
  - Sample variance
- ✓ Standard Deviation
  - Population standard deviation
  - Sample standard deviation

Measures of Dispersion: *Range*

- The Range is the difference between the largest value (Max) and the smallest value (Min).
- Range (R) = Max − Min
- Range =Largest value - Smallest value
- Note: Range concern only onto two values
- Find Range?

Example1: Data: 43, 66, 61, 64, 65, 38, 59, 57, 57, 50.

Example2 Data: 26, 25, 35, 27, 29, 29.

Measures of Dispersion: *Variance*

- The variance is one of the most important measures of dispersion.
- The variance is a measure that uses the mean as a point of reference.
- The variance of the data is small when the observations are close to the mean.
- The variance of the data is large when the observations are spread out from the mean.
- The variance of the data is zero (no variation) when all observations have same value.

Measures of Dispersion: *Standard Deviation*

Standard Deviation

- The Standard Deviation is a measure of how spread out numbers are.
- It is the **square root** of the **Variance.**

Steps for Finding Sample Variance & Sample Standard Deviation

1. Find the mean of the sample data set. $\bar{x} = \frac{\Sigma x}{n}$
2. Find deviation of each entry. $x - \bar{x}$
3. Square each deviation. $(x - \bar{x})^2$
4. Add to get the sum of the deviations squared. $\Sigma(x - \bar{x})^2$
5. Divide by $n - 1$ to get the **sample variance**. $s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$
6. Find the square root to get the **sample standard deviation**. $s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$

Variance and Standard Deviation

Sample Variance, : s^2

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n-1}$$

Sample Standard Deviation, *s:*

$$s = \sqrt{s^2} = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}}$$

Sample versus Population Standard Deviation

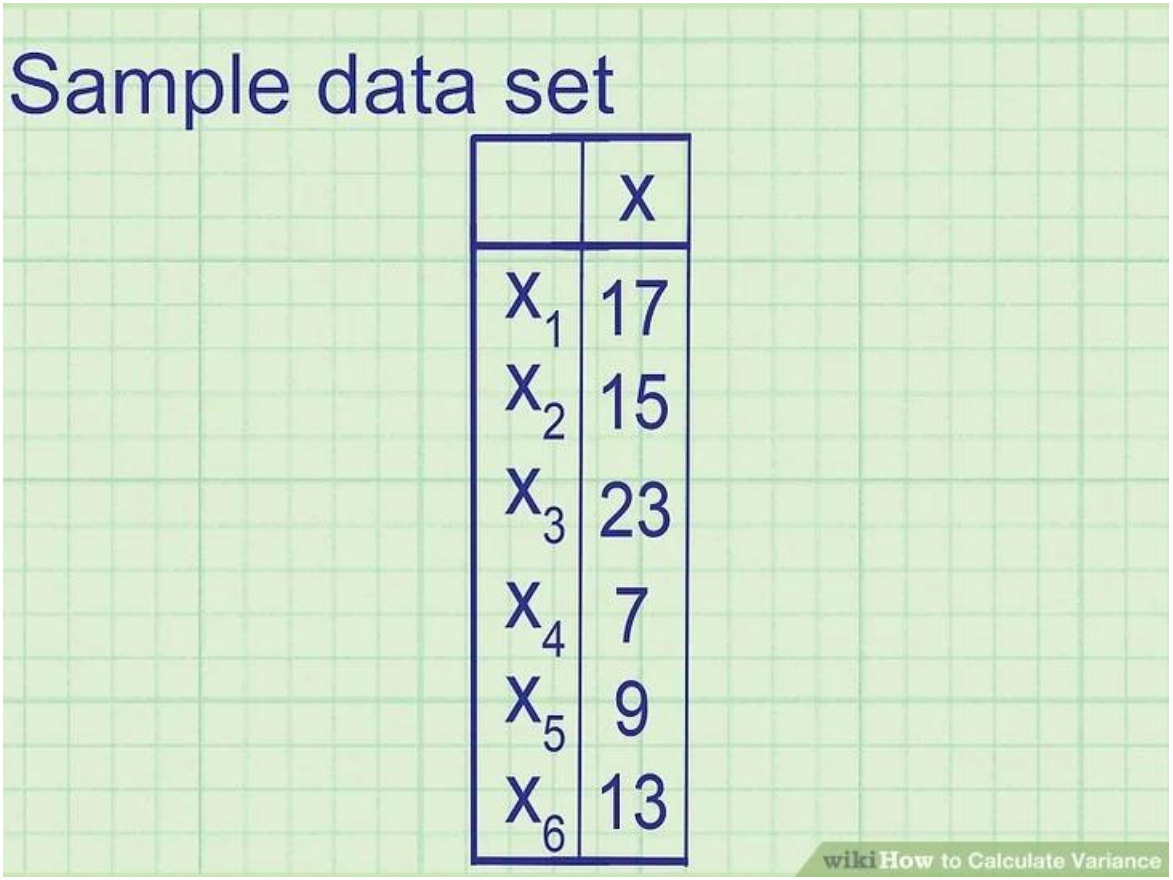
Sample Standard Deviation

- $$s = \sqrt{s^2} = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$

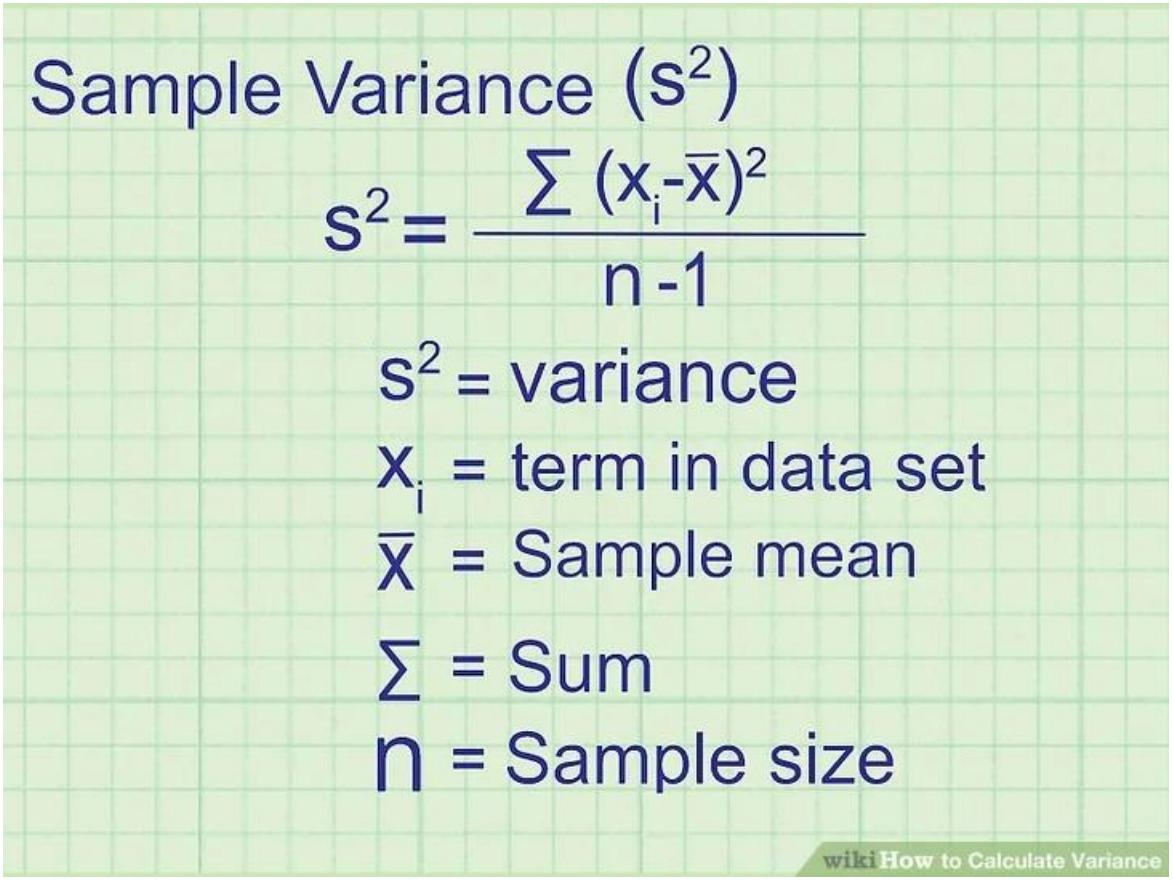
Population Standard Deviation

- $$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\Sigma(x-\mu)^2}{N}}$$

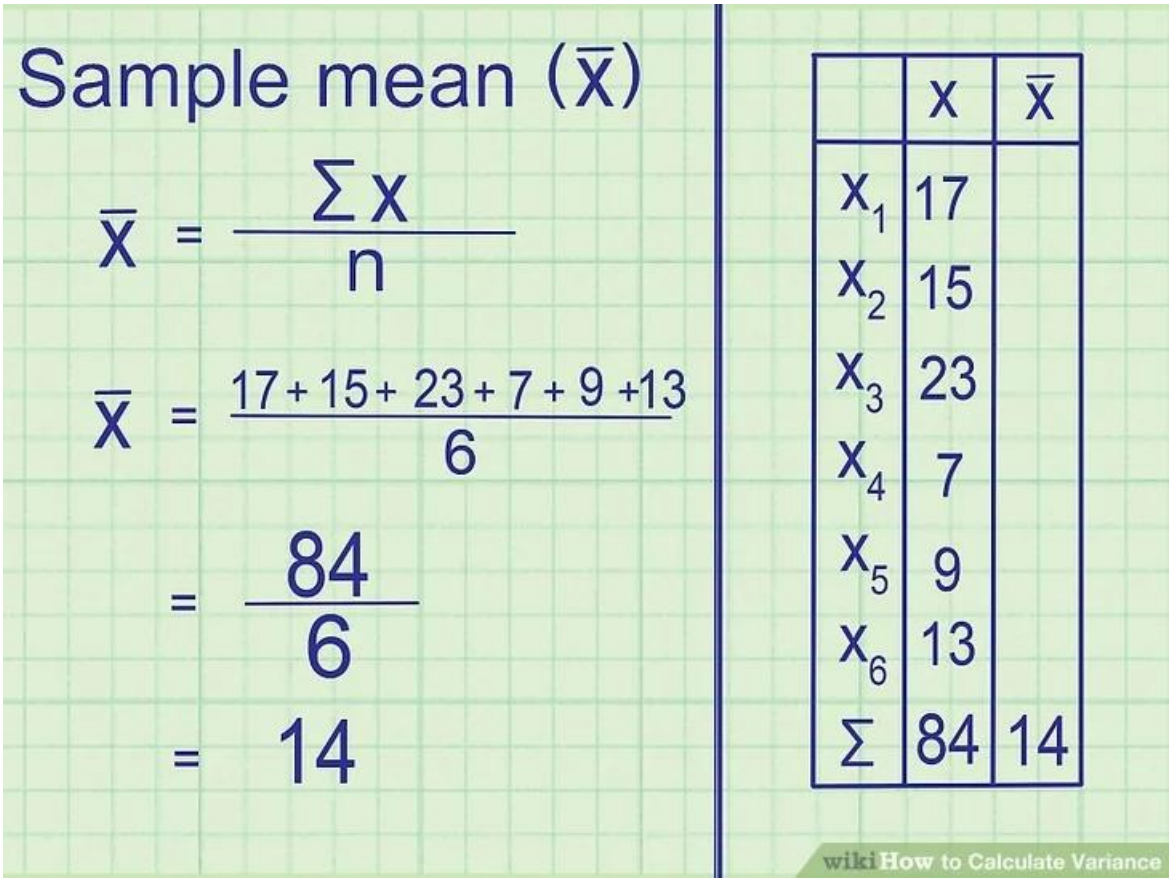
Calculate Sample Variance



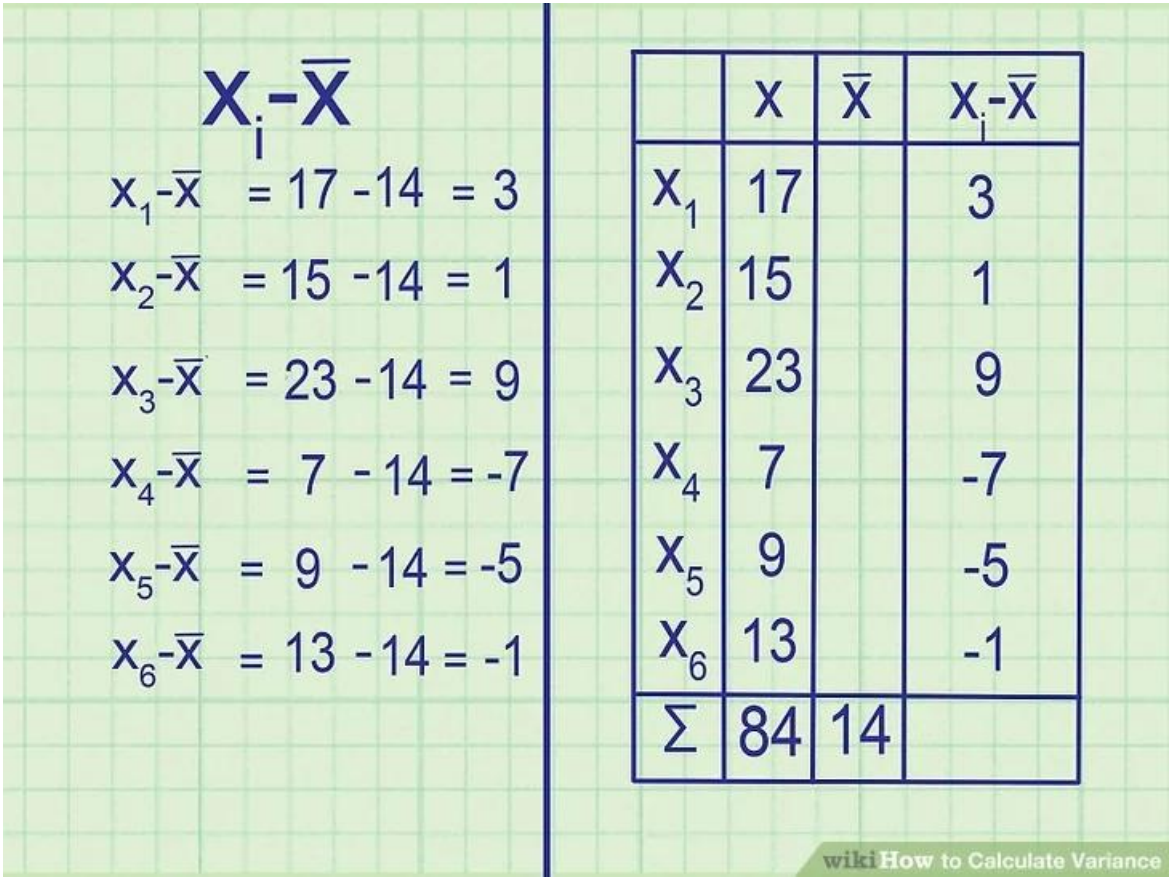
Calculate Sample Variance



Calculate Sample Variance

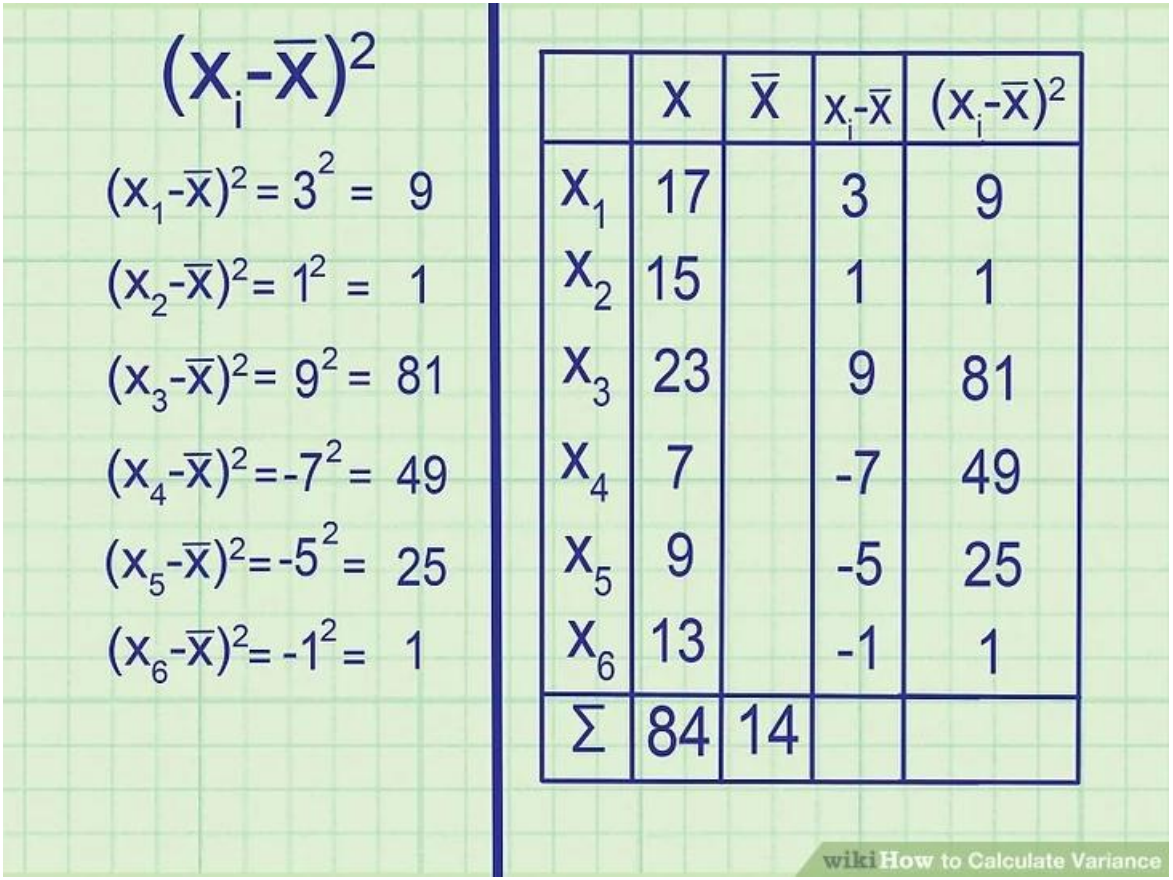


Calculate Sample Variance



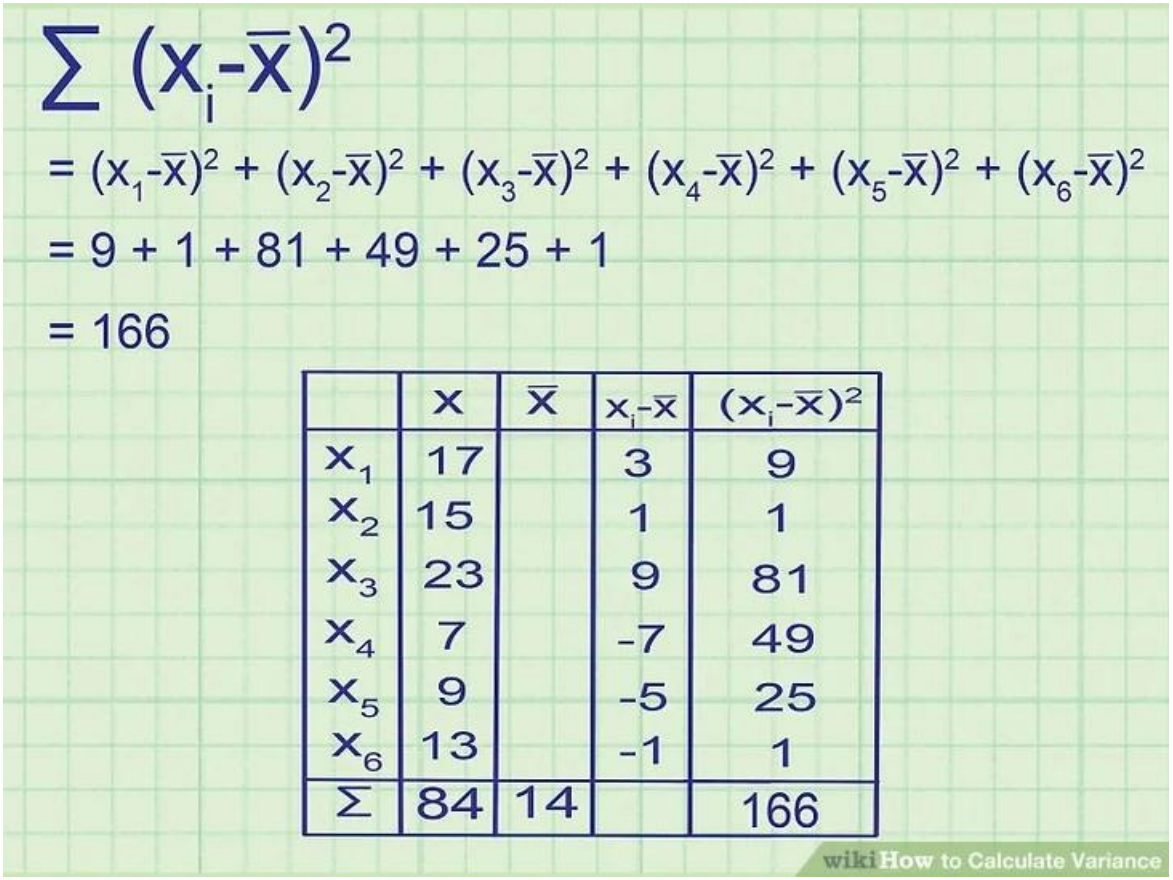
	x	$\bar{x}$	$x_i - \bar{x}$
x_1	17		3
x_2	15		1
x_3	23		9
x_4	7		-7
x_5	9		-5
x_6	13		-1
Σ	84	14	

Calculate Sample Variance



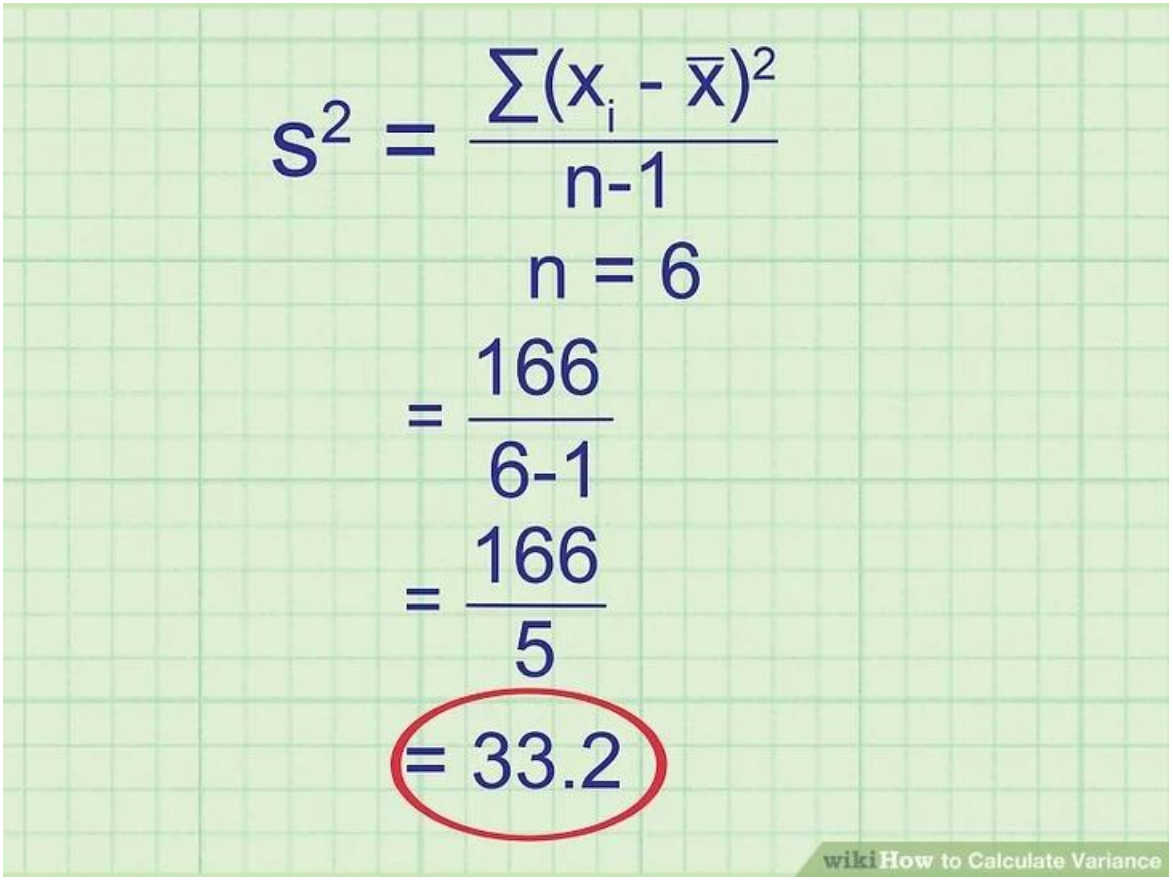
	x	$\bar{x}$	$x_i-\bar{x}$	$(x_i-\bar{x})^2$
x_1	17		3	9
x_2	15		1	1
x_3	23		9	81
x_4	7		-7	49
x_5	9		-5	25
x_6	13		-1	1
Σ	84	14		

Calculate Sample Variance

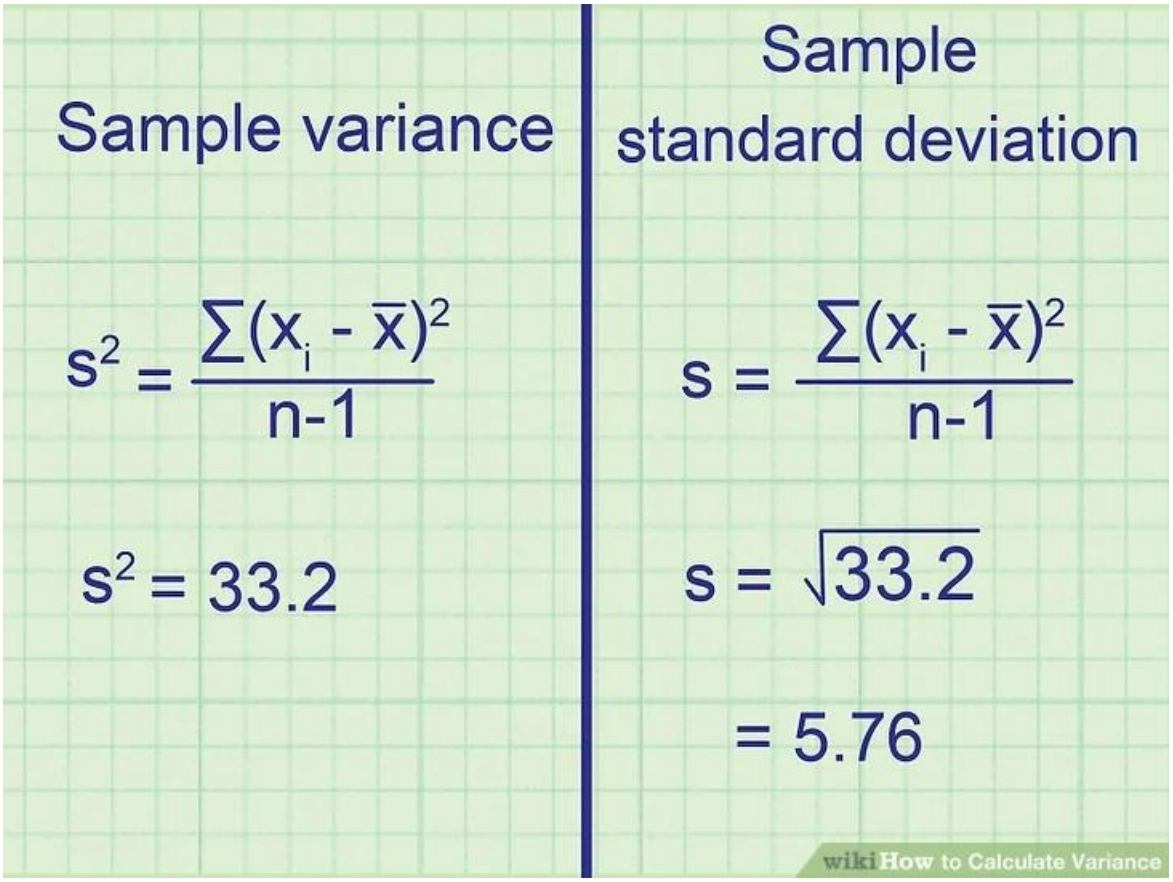


	x	$\bar{x}$	$x_i-\bar{x}$	$(x_i-\bar{x})^2$
x_1	17		3	9
x_2	15		1	1
x_3	23		9	81
x_4	7		-7	49
x_5	9		-5	25
x_6	13		-1	1
Σ	84	14		166

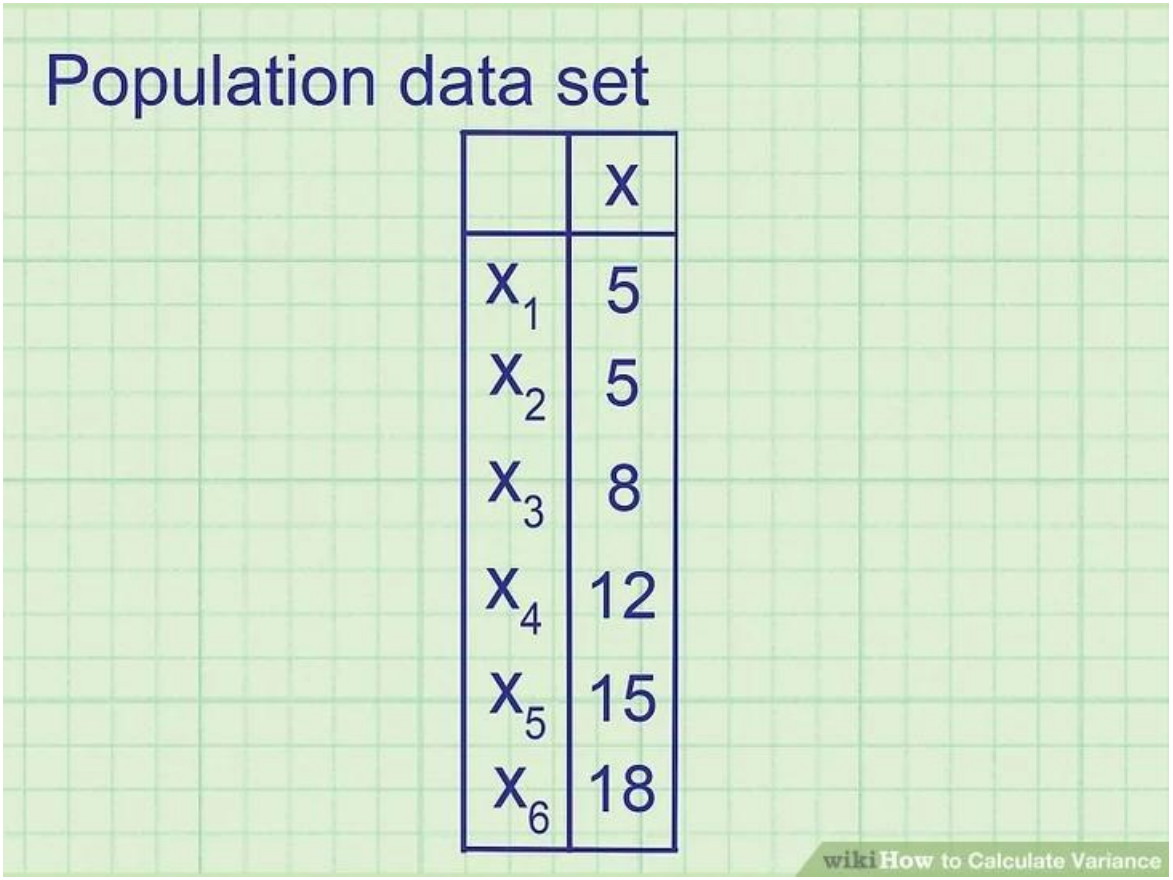
Calculate Sample Variance



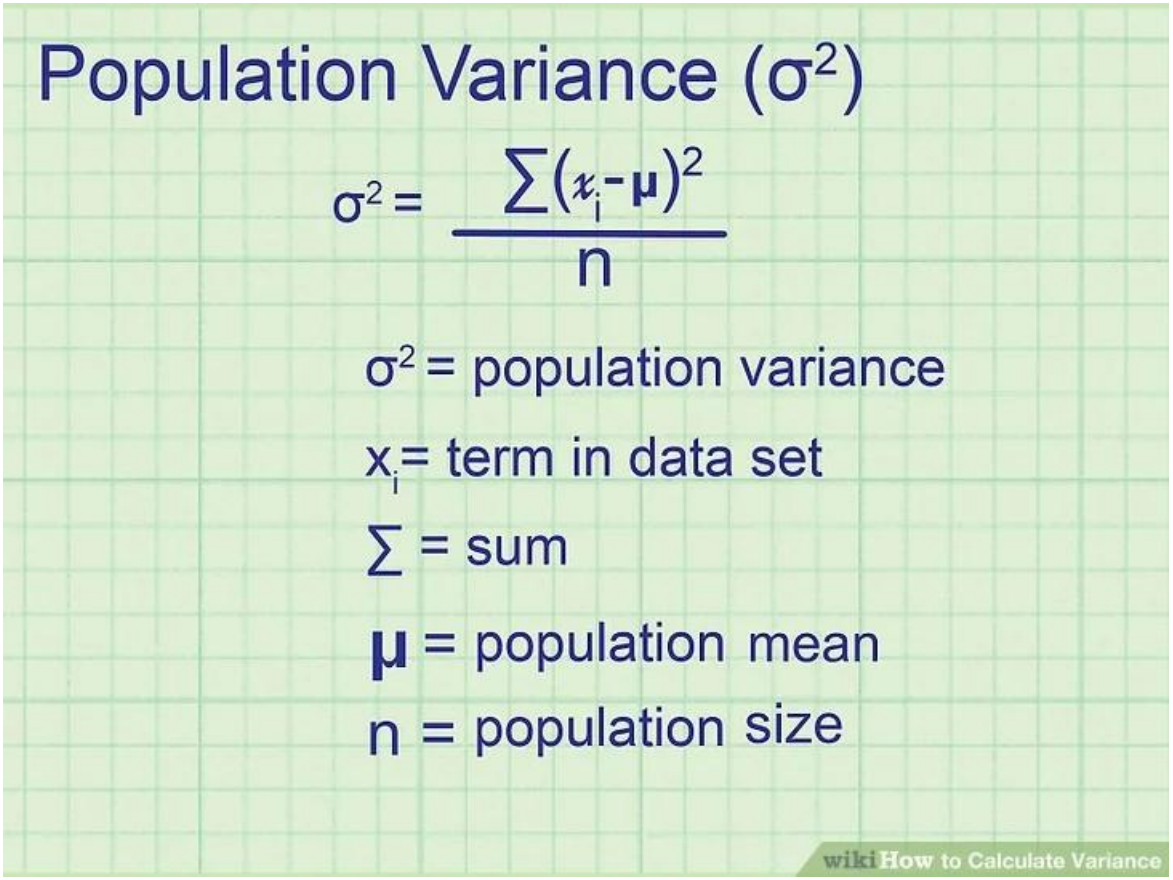
Calculate Sample Variance



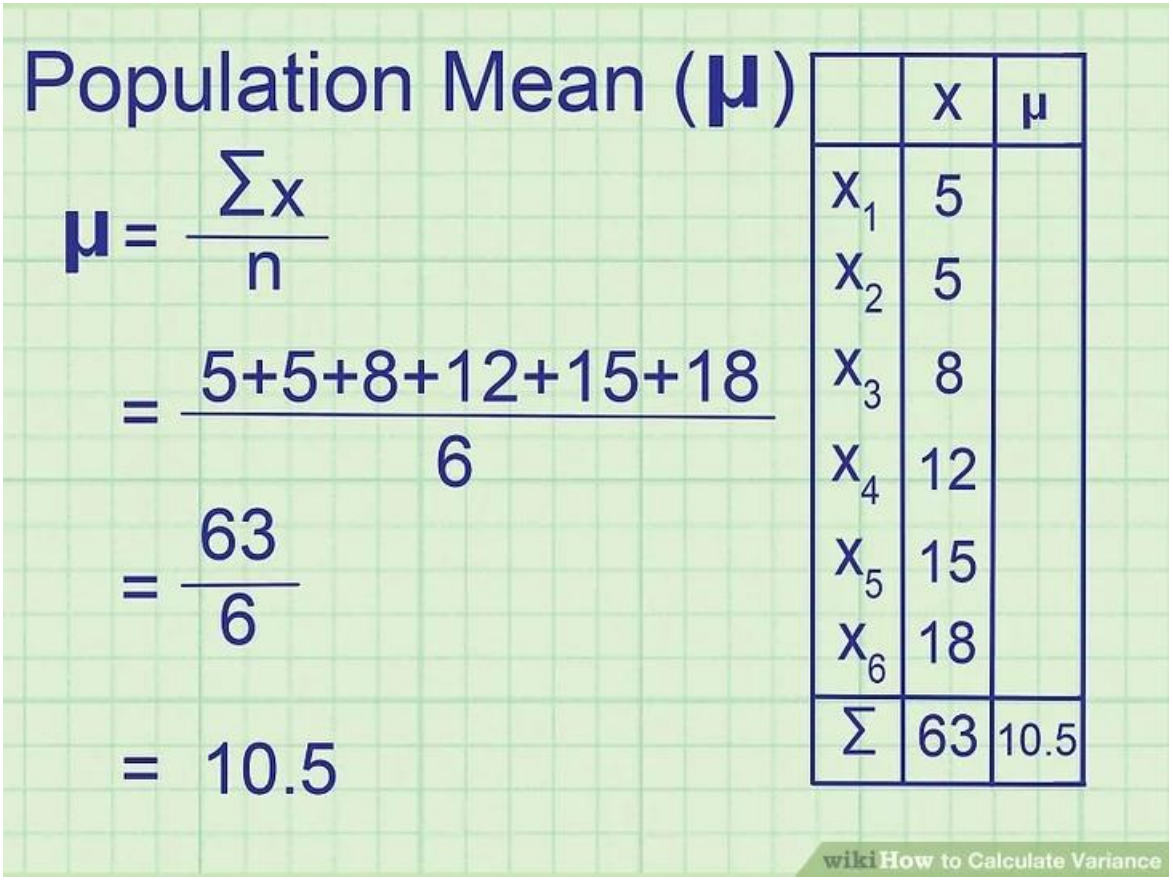
Calculate Population Variance



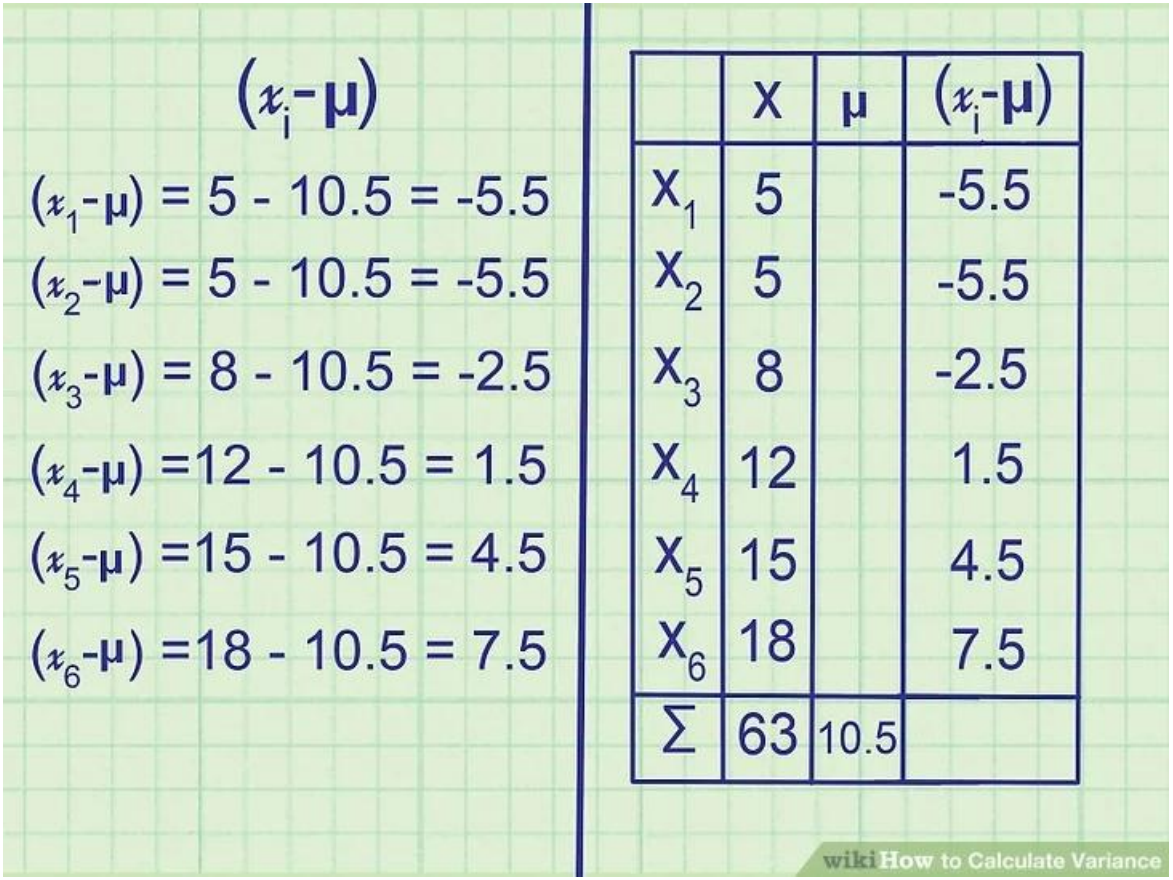
Calculate Population Variance



Calculate Population Variance

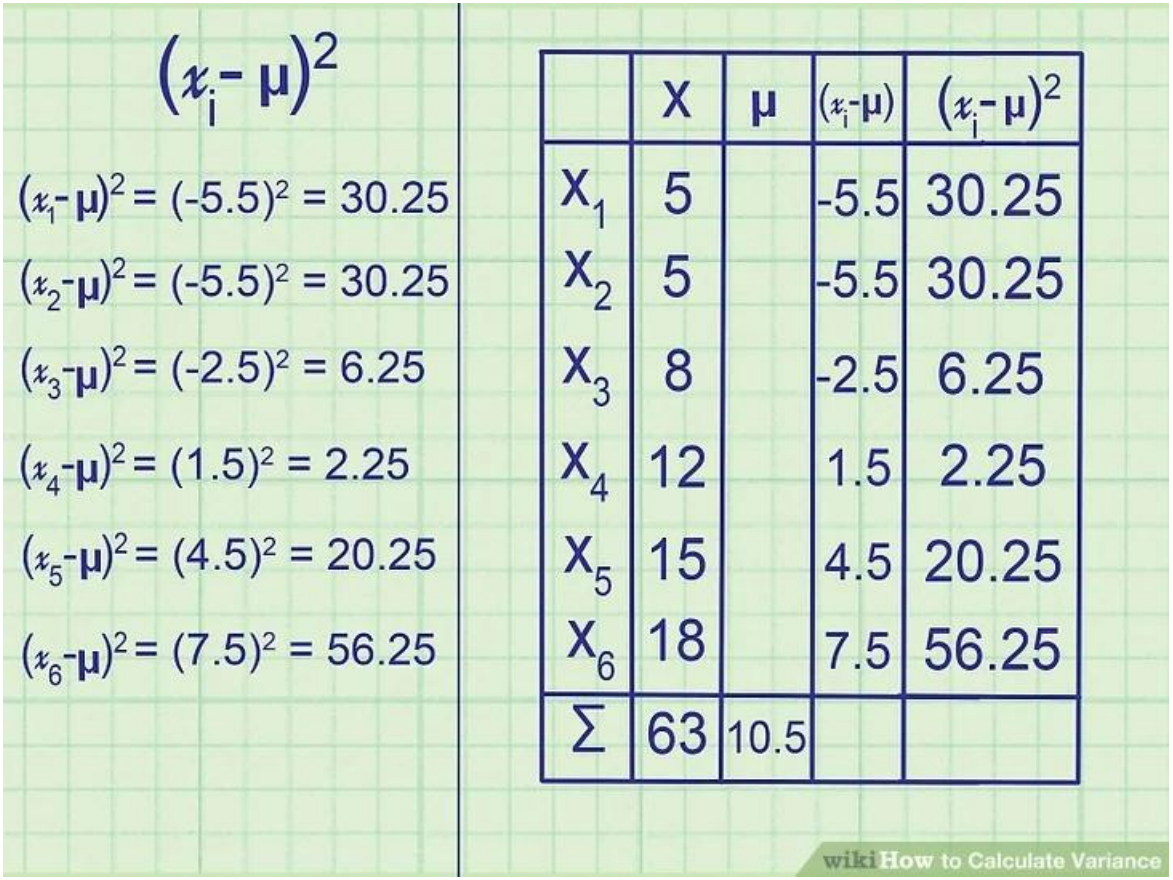


Calculate Population Variance



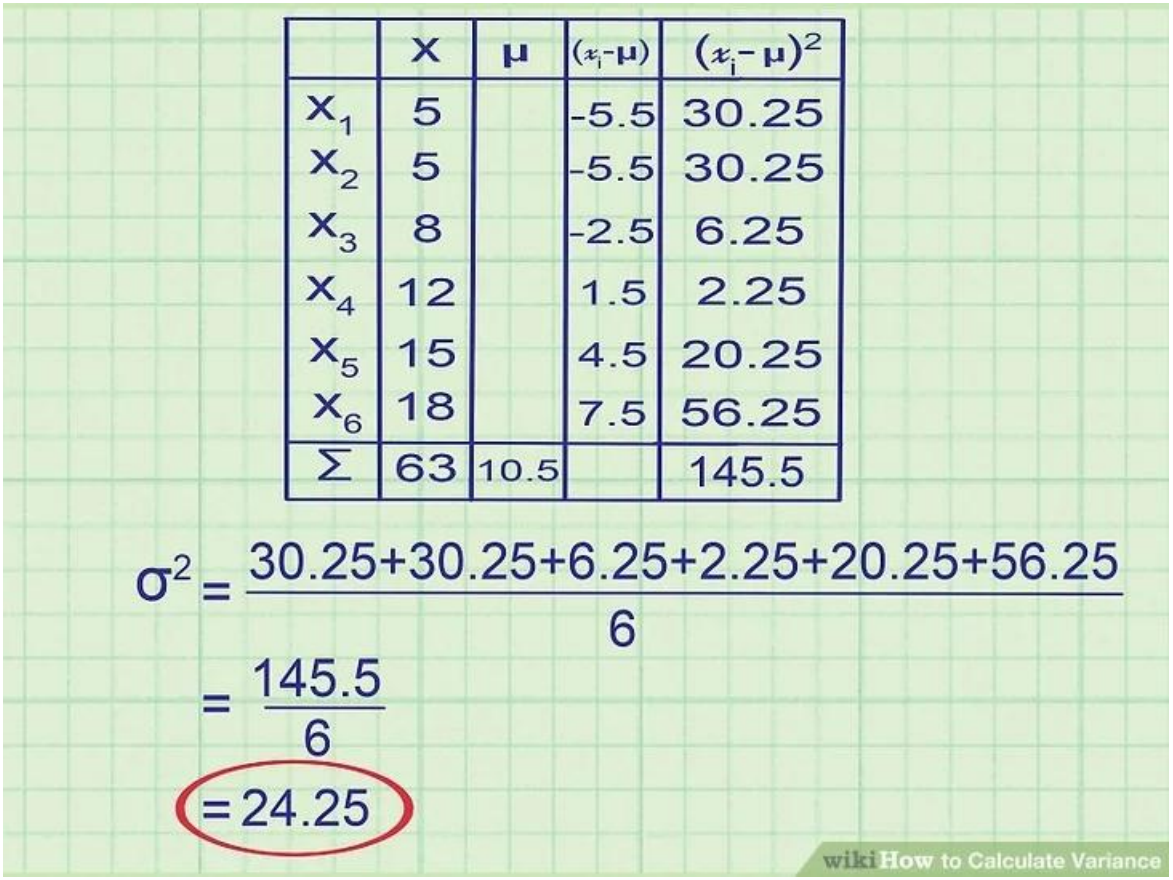
	X	μ	$(x_i-\mu)$
X_1	5		-5.5
X_2	5		-5.5
X_3	8		-2.5
X_4	12		1.5
X_5	15		4.5
X_6	18		7.5
Σ	63	10.5	

Calculate Population Variance



	X	μ	$(x_i-\mu)$	$(x_i-\mu)^2$
X_1	5		-5.5	30.25
X_2	5		-5.5	30.25
X_3	8		-2.5	6.25
X_4	12		1.5	2.25
X_5	15		4.5	20.25
X_6	18		7.5	56.25
Σ	63	10.5		

Calculate Population Variance



THANK YOU